The classic 12-bit FAT filesystem

WHAT IS THIS FILESYSTEM?

FAT12 is the small-volume member of the File Allocation Table family. It stores cluster-chain links in packed 12-bit entries and was widely used on floppy disks.

How the disk is organised

Simplified illustration; not a sector-for-sector specification.

What this implementation provides

- Packed 12-bit allocation entries; fixed-size root directory
- Small floppy-style volumes with fewer than 4,085 data clusters
- V1 reading, V3/V4 filesystem operations and independent Defragmenter interface
- Generic Boot bridge can read the same FAT12.so

One installed module and one manual

```
/System/Filesystems/FAT12.so
/System/Resources/Documentation/
FAT12.pdf
```

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How to use the installed module

The FAT12.so driver is a shared library. It is loaded by a compatible filesystem consumer through its published versioned interfaces; it is not a command to run directly.

Installation and verification

Portable package: install the released Debian .deb.

Native option: run the released .run installer.

The Debian package owns exactly one .so and this PDF.

Build records and checksums live in the Debian package metadata.
Full source, tests and engineering notes remain in GitHub.

Current limits and boundaries

- Large disks and FAT16/FAT32 volumes are different formats.
- Linux .ko and UEFI .efi are separate transitional compatibility assets.
- Formatting and defragmentation can modify data; test on disposable images first.

Operational model

```
Media -> host I/O callbacks -> filesystem .so -> caller
```

The host supplies target selection and protected byte I/O. The filesystem driver interprets that filesystem’s on-disk structures. Ownership, verification and all destructive actions remain the responsibility of the calling application.

This is a user guide, not a claim of untested kernel or firmware support.

Build and installation

Filesystem-FAT12

Release 0.1.10

Installed files

```
/System/Filesystems/FAT12.so

/System/Resources/Documentation/Filesystem-FAT12.pdf

/System/Kernel/Filesystems/fat12.ko
```

Debian package

Install the prebuilt files with dpkg -i followed by the Debian filename.

Manual RUN

Run bash followed by the native installer filename.

The installer compiles the shared object and generates this manual locally.

It also compiles the Linux adapter against the running kernel headers.

Use --kernel-build DIRECTORY to select another prepared kernel build.

The manual installer replaces all three files.

InfiltratorOS installation

The OS installer builds a missing or older shared object.

It verifies the manual independently and repairs it from its matching release.

The kernel module is stored for manual use under its single filename.

The package metadata records the source revision and SHA-256 checksums.